Tutorial 9 Solution

Problem 1

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by $f(x, y) = x - y$. Solve the following problems graphically:

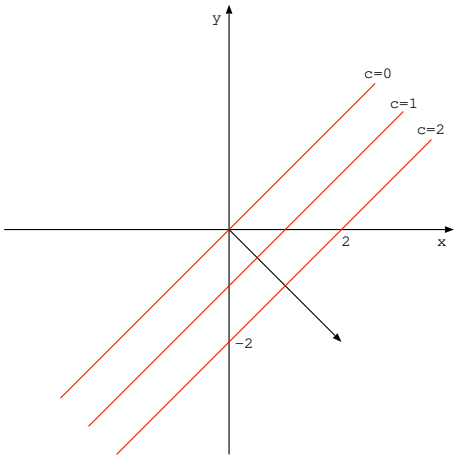
(a) $\max_{x,y\in\mathbb{R}} f(x, y)$

(b) $\max_{x,y\in\mathbb{R}} f(x, y)$ subject to $x^2 + y^2 = 8$

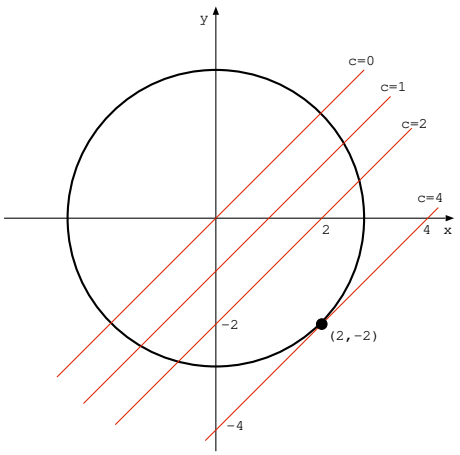
(c) $\max_{x\geq 0,\ y\geq 0} f(x, y)$ subject to $x^2 + y^2 \leq 8$

Solution: In all three cases, the level curve of level c comprises all $(x, y) \in \mathbb{R}^2$ that satisfy $x - y = c$, or, equivalently, $y = x - c$.

(a) We see that there is no maximum — "we can always increase c." That is, the problem has no solution.

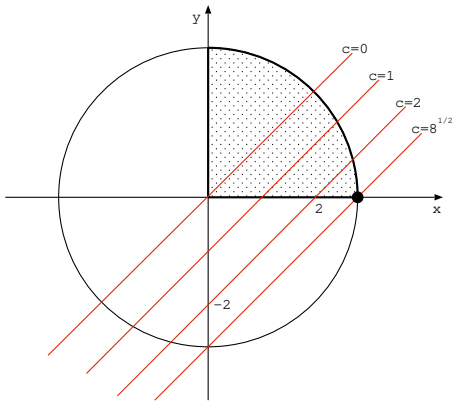


(b) We see that the solution is the tangency point depicted below.



By the implicit function theorem, the slope of the line tangent to the constraint set $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 8\}$ at (x, y) is $-\frac{2x}{2y}$ (as long as $y \neq 0$). Since the slope of the level curves is 1, we must have $-\frac{2x}{2y} = 1$ at every point (x, y) at which a level curve is tangent to the constraint set. Hence $x = -y$ and $x^2 + y^2 = 8$ implies $2x^2 = 8$ and $x = \pm 2$. The maximum is $(2, -2)$ (at this point, the level curve of level $c = 4$ is tangent to the constraint set.) At $(-2, 2)$, f has a minimum in the constraint set.

(c) We see that the maximum is $(\sqrt{8}, 0)$, a point on the level curve of level $c = \sqrt{8}$



Problem 2

A factory makes two kinds of juices: orange juice and apple juice. Each liter of orange juice can be sold for \$1, while each liter of apple juice can be sold for \$2. The production capacity of the factory limits them to producing a total of 2,000 liters of juice. Each liter of orange juice requires 1 hour of labor to process and package. Each liter of apple juice requires 8/3 hours of labor to process and package. The company has a total of 4,000 hours of labor available. Government regulation rations the production of orange juice. The factory has a license that permits it to produce up to 1,500 gallons of orange juice. What mix of orange and apple juice should be produced to maximize the revenue of the factory?

The problem of the factory can be formulated as a linear program. Let x_1 denote the number of liter of orange juice (measured in units of a thousand) to be produced and x_2 be the number of liters of apple juice produced (measured in units of a thousand). Since a non-negative amount must be produced, $x_1, x_2 \geq 0$. Revenue will be $x_1 + 2x_2$. The total amount produced $x_1 + x_2$ must be at most 2, i.e., $x_1 + x_2 \leq 2$. Similarly, the limit on labor time means that $x_1 + (8/3)x_2 \leq 4$. Finally, the government constraint requires that $x_1 \leq 1.5$. Summarizing, the problem of the factory is

$$\begin{aligned} \max \quad & x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + \frac{8}{3}x_2 \leq 4, \\ & x_1 + x_2 \leq 2, \\ & x_1 \leq \frac{3}{2}, \\ & x_1, x_2 \geq 0. \end{aligned}$$

(a) Use the extreme value theorem to determine whether the problem has a solution.

(b) Use the fundamental theorem of linear programming to find a solution to the linear program.

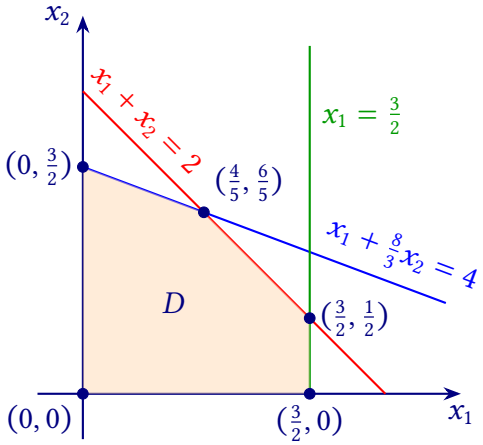
(c) Write down the dual of this linear program.

(d) Interpret the dual variables.

(e) Determine whether the optimal value of the dual is the same as the primal above.

Solution:

(a) Because the objective function is linear, it is continuous. The constraint set D, shaded in orange in the picture below, is a compact set. Then by the extreme value theorem, the linear programming problem has a solution.



(b) By the fundamental theorem of linear programming, since a solution exists by part (a), an extreme point (or vertex or corner) of the constraint set must be a solution. Therefore, it suffices to compare the value of the objective function $f(x_1, x_2) \equiv x_1 + 2x_2$ at the five extreme points: $(0, 0)$, $(0, 3/2)$, $(4/5, 6/5)$, $(3/2, 1/2)$, $(3/2, 0)$. Now

$$f(0, 0) = 0,\ f(0, 3/2) = 3,\ f(4/5, 6/5) = 16/5,\ f(3/2, 1/2) = 5/2,\ f(3/2, 0) = 3/2.$$

Since 16/5 is the largest, a solution of this linear programming problem is $(x_1^*, x_2^*) = (4/5, 6/5)$.

(c) The dual is

$$\begin{aligned} \min\ & 4y_1 + 2y_2 + \frac{3}{2}y_3 \\ \text{s.t.}\ & y_1 + y_2 + y_3 \geq 1, \\ & \frac{8}{3}y_1 + y_2 \geq 2, \\ & y_1 \geq 0, y_2 \geq 0, y_3 \geq 0. \end{aligned}$$

(d) The dual variable y_1 can be interpreted as the marginal value of labor, or more specifically, the relative contribution that one unit of labor makes to the total revenue. Similarly, y_2 and y_3 are the marginal values of total production capacity and production limit of orange juice, respectively.

Alternatively, one can also interpret each dual variables as the imputed cost of using an extra unit of each of the resources.

(e) By the duality theorem, since the primal has a solution by part (a), the optimal value of the primal and the dual must coincide.

Problem 3

Consider the linear programming problem

$$\begin{aligned} \max\ & 2x + 7y \\ \text{s.t.}\ & 4x + 5y \leq 20, \\ & 3x + 7y \leq 21, \\ & x \geq 0, y \geq 0. \end{aligned}$$

(a) Use the fundamental theorem of linear programming to find a solution to the linear program; you can assume that a solution exists.

(b) Write down the dual of the linear program.

(c) Suppose the solution to the dual is $(\lambda_1, \lambda_2) = (0, 1)$. Then if we change 20 to 20.1 and 21 to 20.8. What is the corresponding change in the optimal value of the objective function?

Solution:

(a) The extreme points (or vertices or corners) of the constraint set are $(0, 0)$, $(0, 3)$, $(5, 0)$, and $\left(\frac{35}{13}, \frac{24}{13}\right)$. To find the solution, we evaluate the value of $f(x, y) \equiv 2x + 7y$ at each of the extreme points:

$$f(0,0) = 0, \quad f(0,3) = 21, \quad f(5,0) = 10, \quad f\left(\frac{35}{13}, \frac{24}{13}\right) = \frac{238}{13}.$$

Since 21 is the largest, a solution is $(x^*, y^*) = (0, 3)$.

(b) The dual is

$$\begin{aligned} \min\ & 20\mu_1 + 21\mu_2 \\ \text{s.t.}\ & 4\mu_1 + 3\mu_2 \geq 2, \\ & 5\mu_1 + 7\mu_2 \geq 7, \\ & \mu_1 \geq 0, \mu_2 \geq 0. \end{aligned}$$

(c) Recall that the dual variable can be interpreted as the per unit increase in objective function value from relaxing one of the primal constraints for a sufficiently small amount. Then by changing 20 to 20.1 and 21 to 20.8, the change in the optimal value of the objective function is

$$\lambda_1(20.1 - 20) + \lambda_2(20.8 - 21) = 0(0.1) + 1(-0.2) = -0.2.$$

Problem 4

Write down the dual of the following linear programming problems.

(a)

$$\begin{array}{lll} \max & x_1 + x_2 - 3x_3 & \\ \text{s.t.} & x_1 + 2x_2 - 3x_3 & \leq 4 \\ & 4x_1 + 5x_2 - 9x_3 & \leq 13 \end{array}$$

(b)

$$\begin{array}{lll} \max & x_1 + x_2 - 3x_3 & \\ \text{s.t.} & x_1 + 2x_2 - 3x_3 & = 4 \\ & 4x_1 + 5x_2 - 9x_3 & = 13 \end{array}$$

Solution: Letting

$$\boldsymbol{c} = \begin{pmatrix} 1 & 1 & -3 \end{pmatrix}, \quad \boldsymbol{A} = \begin{pmatrix} 1 & 2 & -3 \\ 4 & 5 & -9 \end{pmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{pmatrix} 4 \\ 13 \end{pmatrix},$$

the linear programs in parts (a) and (b) can be written as

$$\max \boldsymbol{c} \cdot \boldsymbol{x} \quad \text{s.t. } \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

and

$$\max \boldsymbol{c} \cdot \boldsymbol{x} \quad \text{s.t. } \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},$$

respectively. Now using the observations in lecture 9 slides under the "writing down the dual" section, their duals are

$$\min \boldsymbol{y} \cdot \boldsymbol{b} \quad \text{s.t. } \boldsymbol{y}\boldsymbol{A} = \boldsymbol{c},\ \boldsymbol{y} \geq \boldsymbol{0},$$

and

$$\min \boldsymbol{y} \cdot \boldsymbol{b} \quad \text{s.t. } \boldsymbol{y}\boldsymbol{A} = \boldsymbol{c},$$

respectively. Therefore, the dual of the problem in part (a) is

$$\begin{array}{ll} \min & 4y_1 + 13y_2 \\ \text{s.t.} & y_1 + 4y_2 = 1 \\ & 2y_1 + 5y_2 = 1 \\ & -3y_1 - 9y_2 = -3 \\ & y_1 \geq 0,\ y_2 \geq 0 \end{array}$$

and the dual of the problem in part (b) is

$$\begin{array}{ll} \min & 4y_1 + 13y_2 \\ \text{s.t.} & y_1 + 4y_2 = 1 \\ & 2y_1 + 5y_2 = 1 \\ & -3y_1 - 9y_2 = -3 \end{array}$$

Problem 5 (Extra Practice Problem)

An electronics firm that produces phones and tablets has two factories that jointly produce the two goods in the following quantities per hour:

	Factory 1	Factory 2
Phones	10	20
Tablets	25	25

The firm receives an order for 300 phones and 500 tablets. The costs of operating the two factories are \$10000 and \$8000 per hour, respectively.

(a) Formulate the linear programming problem of minimizing the total cost of meeting this order.

(b) Write down the equivalent maximization problem, and then find its dual.

Solution:

(a) Let x_1 and x_2 denote the number of hours for which the two factories operate to produce the order. Then the above table tells us that a total of $10x_1 + 20x_2$ phones is produced, as well as $25x_1 + 25x_2$ tablets. Because 300 phones and 500 tablets are required, the pair (x_1, x_2) must satisfy

$$\begin{aligned} 10x_1 + 20x_2 &\geq 300 \\ 25x_1 + 25x_2 &\geq 500 \end{aligned}$$

In addition, of course, $x_1 \geq 0$ and $x_2 \geq 0$. The total cost of operating the two factories for x_1 and x_2 hours, respectively, is $10000x_1 + 8000x_2$. The problem is, therefore,

$$\min 10000x_1 + 8000x_2 \quad \text{s.t.} \begin{cases} 10x_1 + 20x_2 \geq 300 \\ 25x_1 + 25x_2 \geq 500 \end{cases} \text{and } x_1 \geq 0, x_2 \geq 0$$

(b) The linear programming problem above is equivalent to (also changed $\geq$ to $\leq$ in the constraints)

$$\max -10000x_1 - 8000x_2 \quad \text{s.t.} \begin{cases} -10x_1 - 20x_2 \leq -300 \\ -25x_1 - 25x_2 \leq -500 \end{cases} \text{and } x_1 \geq 0, x_2 \geq 0$$

Its dual is

$$\min -300y_1 + -500y_2 \quad \text{s.t.} \begin{cases} -10y_1 - 25y_2 \geq -10000 \\ -20y_1 - 25y_2 \geq -8000 \end{cases} \text{and } y_1 \geq 0, y_2 \geq 0$$